Preprint

SUPPLEMENTARY MATERIAL

The Spatial Geometry of the Lorentz Tetrad

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Abstract
The legs of a Lorentz tetrad have equal magnitude and are pairwise orthogonal, so the magnitude pattern $|\eta_{ab}| = \delta_{ab}$ of the Minkowski metric is invariant under every permutation of the four legs; only the sign of η_{00} distinguishes the time leg. We realise the four legs as vectors in the three-dimensional spatial sector and ask which configurations are possible. Three premises, each read from η, suffice: the Gram matrix of the realisation inherits the permutation symmetry of $|\eta|$; the legs are distinct; and the realisation lives in $\mathbb{R}^3$. These force the Gram matrix $G = c^2 I - (c^2/3)(J - I)$, and hence the regular tetrahedron, unique up to $O(3)$. Equal norms and closure $\sum_a v_a = 0$ are derived, not assumed. Three is the only dimension in which such a realisation both exists and is forced. A second route, which grants only the symmetry of η itself, gives the same tetrahedron once closure is assumed. Each premise is shown to be necessary. Parity completion gives the cube and the stellated octahedron.

Setting

A Lorentz tetrad has four legs e_a, $a = 0, 1, 2, 3$, with $\eta(e_a, e_b) = \eta_{ab} = \mathrm{diag}(-1, +1, +1, +1)$. Every leg has unit magnitude and every pair is orthogonal, so

$$|\eta_{ab}| = \delta_{ab}, \tag{1}$$

which is invariant under all 24 permutations of the legs. The metric η itself is invariant only under the 6 permutations that fix the time leg; the sign of η_{00} is the only datum that distinguishes it.

A *spatial realisation* is a 4-tuple $V = (v_0, v_1, v_2, v_3)$ of vectors $v_a \in \mathbb{R}^D$, one per leg, with Gram matrix $G_{ab} = v_a \cdot v_b$. For $\sigma \in S_4$ let P_σ be its permutation matrix. We write I for the 4×4 identity, J for the all-ones matrix and $\mathbf{1} = (1, 1, 1, 1)^T$.

Premises

(S) Equivalence of the legs. $P_\sigma G P_\sigma^T = G$ for every $\sigma \in S_4$. A Gram matrix of real vectors is positive semidefinite, so its diagonal cannot carry the sign $\eta_{00} = -1$; what a realisation can inherit from η is its magnitude pattern (1), and with it the permutation symmetry of the legs.

(N) Distinct legs. The v_a are not all equal, as the legs of a tetrad are linearly independent ($\det e \neq 0$).

(D) Dimension. $D = 3$: the realisation lives in the spatial sector, the three positive entries of η.

Neither equal norms nor closure is assumed.

Two Lemmas

Lemma 1 (commutant). *G satisfies (S) if and only if $G = \alpha I + \beta J$ for some $\alpha, \beta \in \mathbb{R}$.*

Proof. (S) states $G_{\sigma(a)\sigma(b)} = G_{ab}$ for all σ, a, b, so G is constant on the orbits of S_4 on ordered pairs. There are two: the four diagonal pairs, and the twelve off-diagonal pairs, one orbit because S_4 is 2-transitive. Writing the two values as $\alpha + \beta$ and β gives $G = \alpha I + \beta J$; conversely I and J commute with every P_σ. □

Lemma 2 (rigidity). *If $U, W \in \mathbb{R}^{D \times 4}$ have $U^T U = W^T W$ of rank D, then $W = QU$ for a unique $Q \in O(D)$.*

Proof. For every $\lambda \in \mathbb{R}^4$, $|U\lambda|^2 = \lambda^T U^T U \lambda = |W\lambda|^2$. Hence $U\lambda = U\lambda'$ implies $W\lambda = W\lambda'$, and $Q : U\lambda \mapsto W\lambda$ is a well-defined linear isometry from the column space of U onto that of W. Both are $\mathbb{R}^D$ because the rank is D, so $Q \in O(D)$, and it is fixed by its values on the columns. □

The Forced Gram Matrix

Theorem. *Under (S), (N) and (D), the legs have a common norm $c > 0$, they close, $\sum_a v_a = 0$, and*

$$\boxed{G = c^2 I - \frac{c^2}{3}(J - I),} \tag{2}$$

that is $G_{aa} = c^2$ and $G_{ab} = -c^2/3$ for $a \neq b$.

Proof. By Lemma 1, $G = \alpha I + \beta J$. Since $J\mathbf{1} = 4\mathbf{1}$ and J vanishes on $\mathbf{1}^\perp$, G has eigenvalue $\alpha + 4\beta$ on $\mathbf{1}$ and α, three times, on $\mathbf{1}^\perp$. Every pair of legs is at the same distance, $|v_a - v_b|^2 = G_{aa} + G_{bb} - 2G_{ab} = 2\alpha$, so by (N) $\alpha > 0$ and $\mathrm{rank}\, G \geq 3$. By (D), $\mathrm{rank}\, G \leq 3$. Hence $\alpha + 4\beta = 0$, so $G\mathbf{1} = 0$ and

$|\sum_a v_a|^2 = \mathbf{1}^T G \mathbf{1} = 0$: the legs close. Their common norm is $c^2 = G_{aa} = \alpha + \beta = 3\alpha/4$, and $\beta = -\alpha/4 = -c^2/3$, which is Eq. (2). □

Corollary 1 (the regular tetrahedron). *The four points v_a are pairwise equidistant, with edge $c\sqrt{8/3}$ and centroid at the origin: they are the vertices of a regular tetrahedron of circumradius c. With $c = \sqrt{3}$ the columns of*

$$M = \begin{pmatrix} +1 & +1 & -1 & -1 \\ +1 & -1 & +1 & -1 \\ +1 & -1 & -1 & +1 \end{pmatrix} \tag{3}$$

realise Eq. (2), and every realisation is $V = (c/\sqrt{3})\,QM$ for a unique $Q \in O(3)$.

Proof. The distances are $2\alpha = 8c^2/3$ and closure puts the centroid at the origin. Each column of M has three entries ± 1, and each pair of columns agrees in exactly one place, so $M^T M = 3I - (J - I)$. Both V and $(c/\sqrt{3})M$ have Gram matrix (2), of rank 3, and Lemma 2 gives Q. □

Corollary 2 (why three dimensions). *For K legs satisfying (S) and (N) in $\mathbb{R}^D$, a realisation exists if and only if $D \geq K - 1$, and it is forced to the closed regular simplex if and only if $D = K - 1$. For the tetrad, $K = 4$ and $D = 3$.*

Proof. The argument above with K in place of 4 gives eigenvalues $\alpha > 0$ ($K - 1$ times) and $\alpha + K\beta$, so $\operatorname{rank} G \geq K - 1$ and no realisation exists for $D < K - 1$. The regular simplex lies in $\mathbb{R}^{K-1} \subseteq \mathbb{R}^D$, so one exists for $D \geq K - 1$. If $D = K - 1$, then $\alpha + K\beta = 0$ and the simplex is forced. If $D \geq K$, any $\alpha + K\beta > 0$ is also realisable, for example $\beta = 0$: K orthogonal legs of equal length. □

Corollary 3 (parity completion). *Spatial parity $v_a \to -v_a$ preserves G, so $-V$ is a second realisation. V and $-V$ share no point, and for $V = M$ their eight points are the vertices of the cube $\{-1, +1\}^3$; the compound of the two tetrahedra is Kepler's stella octangula [1, 2], the stellated octahedron.*

Proof. $(-V)^T(-V) = V^T V$. A shared point $v_a = -v_b$ would need $G_{ab} = -c^2$, but $G_{ab} \in \{c^2, -c^2/3\}$. The columns of M and $-M$ are eight distinct sign triples, which exhaust $\{-1, +1\}^3$. □

A Second Route

A reader may grant only the symmetry of η itself, the permutations of the three spatial legs, together with equal norms. Closure must then be assumed, and the result is the same.

Proposition. *Let G be invariant under the permutations of legs $1, 2, 3$, with $G_{aa} = c^2 > 0$ for all a, and let $\sum_a v_a = 0$. Then G is Eq. (2).*

Proof. The invariance gives $G_{0i} = q$ and $G_{ij} = t$ for $i \neq j$ in $\{1, 2, 3\}$. Closure gives $G\mathbf{1} = 0$. Row 0 reads $c^2 + 3q = 0$, and row i reads $q + c^2 + 2t = 0$, so $q = t = -c^2/3$. □

The time leg, which η singles out by its sign, ends up geometrically equivalent to the spatial legs: the realisation recovers the full permutation symmetry of $|\eta|$.

Necessity of the Premises

Removing any one premise admits configurations that satisfy the others but are not the regular tetrahedron:

Removed	Admitted configurations
(S)	$(v, w, -v, -w)$ with $\lvert v\rvert = \lvert w\rvert$, $v \perp w$: distinct legs in $\mathbb{R}^3$, equal norms and even closed, but a square
(N)	$v_a = v$ for all a: $G = c^2 J$
(D)	In $\mathbb{R}^4$: $v_a = c\,\hat{e}_a$, $G = c^2 I$, the pattern $\lvert\eta\rvert$ itself, not closed. In $\mathbb{R}^2$: no realisation
Closure	(second route) $v_i = c\,\hat{e}_i$, $v_0 = -c(1,1,1)/\sqrt{3}$: $G_{0i} = -c^2/\sqrt{3}$, $G_{ij} = 0$

In the second route, invariance under the spatial legs with equal norms leaves, for distinct legs in $\mathbb{R}^3$, the one-parameter family $c^2(c^2 + 2t) = 3q^2$; closure selects $q = t = -c^2/3$ from it.

Summary

Everything is read from the Minkowski metric. Its magnitude pattern $|\eta_{ab}| = \delta_{ab}$ makes the legs equivalent, and its three positive entries fix the dimension of the spatial sector. In that sector, four equivalent and distinct legs have exactly one configuration up to $O(3)$: the regular tetrahedron, with common norm and closure as consequences. Three is the only dimension in which this configuration both exists and is forced. Parity, which the Gram matrix cannot see, supplies the second tetrahedron, and the two together form the stellated octahedron.

References

[1] J. Kepler, *Harmonices Mundi* (Linz, 1619).

[2] P. R. Cromwell, *Polyhedra* (Cambridge University Press, Cambridge, 1997).